Biometric Matching

Biometric Matching Process

Template Alignment Techniques

- Critical Challenge: Biometric samples vary due to position, rotation and scale changes
  - Brute Force Search
    - Try multiple alignments and select best match
    - Computationally expensive but thorough
    - Used when other methods fail
  - Reference Point Alignment
    - Detect stable reference point(core point in fingerprint, pupil in iris)
    - Align templates based on reference point
    - Fast but depends on accurate detection
  - Iterative Closest Point(ICP)
    - Initialize transformation
    - Find nearest neighbors between templates
    - Estimate optimal transformation
    - Apply transformation
    - Repeat until convergence
  - RANSAC(Random Sample Consensus)
    - Robust to outliers
    - Randomly sample correspondences and find best transformation

Score Normalization Techniques

- Different matchers produce scores on different scales. Normalization is essential for
    - Multimodal fusion
    - Comparison across systems
    - Quality-based weighting

Matching Performance Optimization

1. Indexing and Binning Reduce search space for 1:N identification

- **Fingerprint:** Classify by pattern type (loop, whorl, arch)
- **Face:** Gender, age, ethnicity filters
- **Iris:** Texture-based clustering

2. Cascaded Matching

Step 1: Quick coarse matching to filter candidates

Step 2: Detailed fine matching on filtered subset

Example: Use global features first, then local minutiae details

3. Hardware Acceleration

- GPU parallel processing for deep learning
- FPGA for real-time iris matching
- SIMD instructions for correlation

Quality–based matching: Incorporate quality scores to improve matching decisions

- Image Quality
  - Resolution
  - Contrast
  - Focus/sharpness
  - Lighting
  - Noise level
- Feature Quality
  - Number of features
  - Feature reliability
  - Occlusions
  - Pose variations
  - Expression (face)

Quality-Weighted Matching:

Formula: Final_Score = Match_Score × Quality_Factor

Strategy:

- Low quality sample → Request re-capture
- Medium quality → Adjust threshold
- High quality → Use standard threshold

Comparative Analysis of Matching methods

Method	Accuracy	Speed	Memory	Best For
Correlation	High	Slow	High	Small databases, high security
Minutiae (FP)	High	Fast	Low	Fingerprint systems
Hamming (Iris)	Very High	Very Fast	Very Low	Large-scale iris recognition
Eigenfaces	Medium	Fast	Medium	Controlled face recognition
Deep Learning	Very High	Fast*	High	Modern face, fingerprint systems
DTW (Voice)	Medium-High	Medium	Medium	Voice verification

- After feature extraction; GPU required for real-time performance

Challenges in Biometric matching

- Intra-Class Variations(Same person, different samples)
  - Sensor noise
  - Environmental changes
  - User interaction variability
  - Aging effects
  - Injuries or conditions
  - Pose and expression

- Inter-Class Variations
  - Twins or relatives
  - Limited feature space
  - Population statistics
  - Similar appearance
  - Partial data overlap

Other Challenges:

- **Non-linear distortions:** Elastic deformation in fingerprints, facial expressions
- **Occlusions:** Glasses, masks, jewelry, scars
- **Scalability:** Matching speed degrades with database size
- **Template aging:** Performance decreases over time

Practical Example: Fingerprint matching

Step-by-Step Process

Input: Query fingerprint image, Database of templates

Step 1: Enhancement

- Apply Gabor filters for ridge enhancement
- Binarization and thinning

Step 2: Minutiae Extraction

- Detect ridge endings: count neighboring pixels
- Detect bifurcations: count ridge branches
- Extract (x, y, θ) for each minutia

Step 3: Alignment

- Detect reference point (core/delta)
- Rotate and translate query to align with template
- Try multiple alignment hypotheses if needed

Step 4: Matching

- For each template minutia, find nearest query minutia
- Count matches within tolerance (Δx, Δy, $\Delta\theta$)
- Typical tolerance: 10 pixels, 15 degrees

Step 5: Scoring Match_Score = Matched_Minutiae / min(Query_Minutiae, Template_Minutiae)

Step 6: Decision

- If Match_Score $\geq$ 0.65: MATCH
- Else: NO MATCH

Practical Example: Iris matching

Input: Query iris image, Stored iris code (2048 bits)

Step 1: Preprocessing

- Detect pupil and iris boundaries (circular Hough transform)
- Normalize iris region to fixed dimensions (512×64)
- Compensate for eyelid/eyelash occlusions

Step 2: Feature Encoding

- Apply 2D Gabor wavelets at multiple scales
- Phase quantization: 2 bits per pixel
- Generate 2048-bit iris code
- Create noise mask for occluded regions

Step 3: Matching

- Shift query code ±15 pixels to compensate for rotation
- For each shift position:
  - XOR_result = Query_Code XOR Template_Code
  - Valid_bits = bits NOT masked by noise
  - HD = count(XOR_result) / Valid_bits

Step 4: Decision

- HD = minimum Hamming distance across all shifts
- If HD < 0.32: MATCH (genuine)
- If HD > 0.45: NO MATCH (impostor)
- If 0.32 ≤ HD ≤ 0.45: UNCERTAIN (request re-capture)

Evaluating Matching Performance

- Genuine Score Distribution
  - Matching score for same person
  - Should be high (Cluster near 1.0)

- Imposter Score Distribution
  - Matching score for different people
  - Should be low (Cluster near 0.0)

0 – 0.25

Distribution Separation:

d-prime (d'): Measure of separation between distributions

Formula: $d' = |\mu_genuine - \mu_impostor| / \sqrt{((\sigma_genuine^2 + \sigma_impostor^2) / 2)}$

Higher d' = Better matcher (less overlap)

Decidability Index: Similar to d-prime but uses standard deviations directly

Formula: $D = |\mu_genuine - \mu_impostor| / \sqrt{(\sigma_genuine^2 + \sigma_impostor^2)}$

Security in Biometric matching: Template Protection Schemes

- Cancellable Biometrics
  - Apply one-way transformation
  - Can revoke and reuse
  - Matching in transformed domain
  - **Example**: Random projection, bio-hashing

- Fuzzy Vault
  - Hide template in Chaff points
  - Genuine match unlocks vault
  - Imposter cannot distinguish real from chaff
  - **Used for**:
    - Fingerprint minutiae

Homomorphic Encryption: Perform matching on encrypted templates without decryption

- Cloud-based matching with privacy
- Computationally expensive
- Active research area

Key Principle: Never store or transmit raw biometric data. Always use protected templates.

Implementation Guidelines

- Choosing a Matching Algorithm

Consideration	Questions	Impact
Accuracy Requirements	What FAR/FRR is acceptable?	Choose method with proven performance
Speed Requirements	Real-time vs. batch processing?	Consider computational complexity
Database Size	100s? 1000s? Millions?	Scalability and indexing needs
Hardware Constraints	Embedded? Mobile? Server?	Memory and processing power
Standards Compliance	Industry standards required?	Use standardized formats and methods

Best practices for Biometric Matching

Step 1: Always normalize scores before fusion or comparison

Step 2: Implement quality checks - reject poor quality samples early

Step 3: Use multiple enrollment samples for robust templates

Step 4: Set thresholds based on application requirements, not defaults

Step 5: Log match scores and decisions for performance monitoring

Step 6: Test with diverse population representing actual users

Step 7: Implement liveness detection to prevent spoofing

Step 8: Provide clear feedback to users for failed matches